

Basic Nutrition

Nutrition through life cycle

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- Introduction
- Principles
- Standards
- Carbohydrate
- Protein
- Fat
- Others
- Nutritional assessment
- Pyramids

Refreshing

- **Nutrients:**
- All help build cells and tissues, regulate bodily processes such as breathing.
- Deficiency Disease: failure to meet your nutrient needs.
- **Nutrition** (medically):

- **Food security:**
- **Food insecurity:**
- **Essential nutrients:**
- **Nonessential nutrients: -----**

Life cycle

- Preconception.
- Fetal growth.
- Pregnancy
- Infant.
- Lactation.
- Toddler.
- Preschool.
- Child.
- Preadolescent.
- Adolescent.
- Adult.
- Elderly.
- Old elderly.

- Healthy individuals
- Nutritional needs
- Nutritional status

Standards

DRIs

- Dietary References Intakes .
- Intake of a nutrient.

RDAs

- Recommended Dietary Allowances.

EARs

- Estimated Average Requirements.
- The nutrient intake value that is estimated to reach the requirements in 50 % of people in a specific group,

ULs

- Tolerable Upper Intake Levels.
- Upper healthy intake of a nutrient.

The six classification of nutrients

1. Carbohydrates
2. Proteins
3. Fats
4. Vitamins
5. Minerals
6. water

Carbohydrates

Carbohydrates

- The **body's**
- Each type of simple and complex CHO
- **except** fiber, provides 4 calories per 1 gram.

Glycemic index of CHO

- It was known that CHO is a CHO is CHO.
- It was thought that all types of CHO had the same effect on blood glucose levels and health,

- This is important to people with disorders such as:
 - Insulin resistance
 - Type 2 diabetes

- GI compares the potential of foods containing the same amount of carbohydrate to raise blood glucose.

- Low GI diet

Recommended intake level

- Based on their contribution to total energy intake.

Protein

Proteins

- The major structural molecules in living things for growth and repair : muscles, bones, hair, skin, nails...

- Body can also use protein as a source of energy BUT **it is not the primary** function of protein.
- Provides 4 kcal per 1 gram

Recommended intake

- 10- 35 % of total energy
- Deficiency leads to

Lipids (fats)

LIPIDS

- They are a great source of STORED ENERGY so we have it in the future.

- Fatty acids found in animal triglycerides are saturated ones

EFA

- 2 EFA:

- Saturated and unsaturated fats.
- Hydrogenated and trans-fats.

Recommended intake

- In the past it was 30% of total calories

ENERGY

1 gram	Number of Calories it provides
Carbohydrates	
Proteins	
Lipids	

OTHER SUBSTANCES

Vitamins

- Are complex substances
 - Normal growth, maintenance, and reproduction
 - Your body cannot produce all vitamins.

Fat-Soluble Vitamins

- Vitamin A
- Vitamin D
- Vitamin E
- Vitamin K

Water-Soluble Vitamins

- Vitamin B-Complex
- Vitamin C

- Excessive consumption of the fat-soluble vitamins from supplements, especially vitamin A and D produces toxicity symptoms.
- High intake of water-soluble vitamins from supplements can also produce adverse health effects.

Minerals

- The body needs 15 minerals that help regulate cell function and provide structure for cells.
- **Major** minerals, in terms of amount present, include calcium, phosphorus, and magnesium.

Water

- The most important nutrient,
- involved in every bodily function,
- makes up 70- 75% of total body weight.
- Water helps to

- Divide your body weight by 2, that's about how much water you need each day in ounces.
- 1 ounce= 0.03 litter

Variables which affect nutrient needs:

- 1.

Steps in the development of nutrient deficiency and toxicities

Deficiency

1. Inadequate intake
2. Depletion of tissue reserves of the nutrient
3. Decreased blood level

Toxicity

1. Excessive nutrient intake
2. Saturation of tissue reserves of the nutrient
3. Increased blood level
4. Excessive nutrient available to cells

NUTRITIONAL ASSESSMENTS

- The nutritional status of an individual is often the result of many inter-related factors.

The purpose of nutritional assessment is to:

- Identify individuals or population groups at risk of becoming mal-nourished OR over-nourished

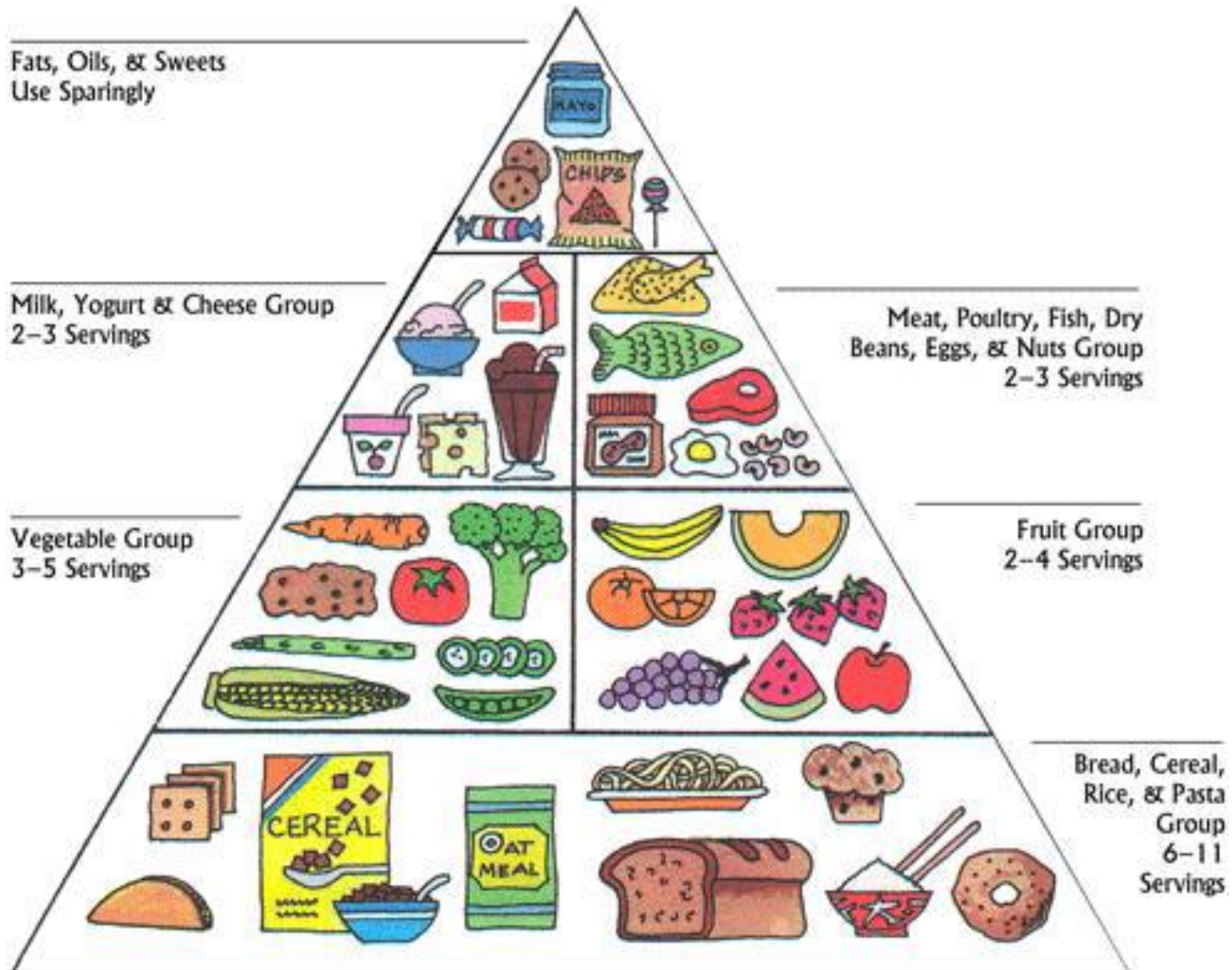
Methods of Nutritional Assessment

Nutrition is assessed by two types of methods; direct and indirect.

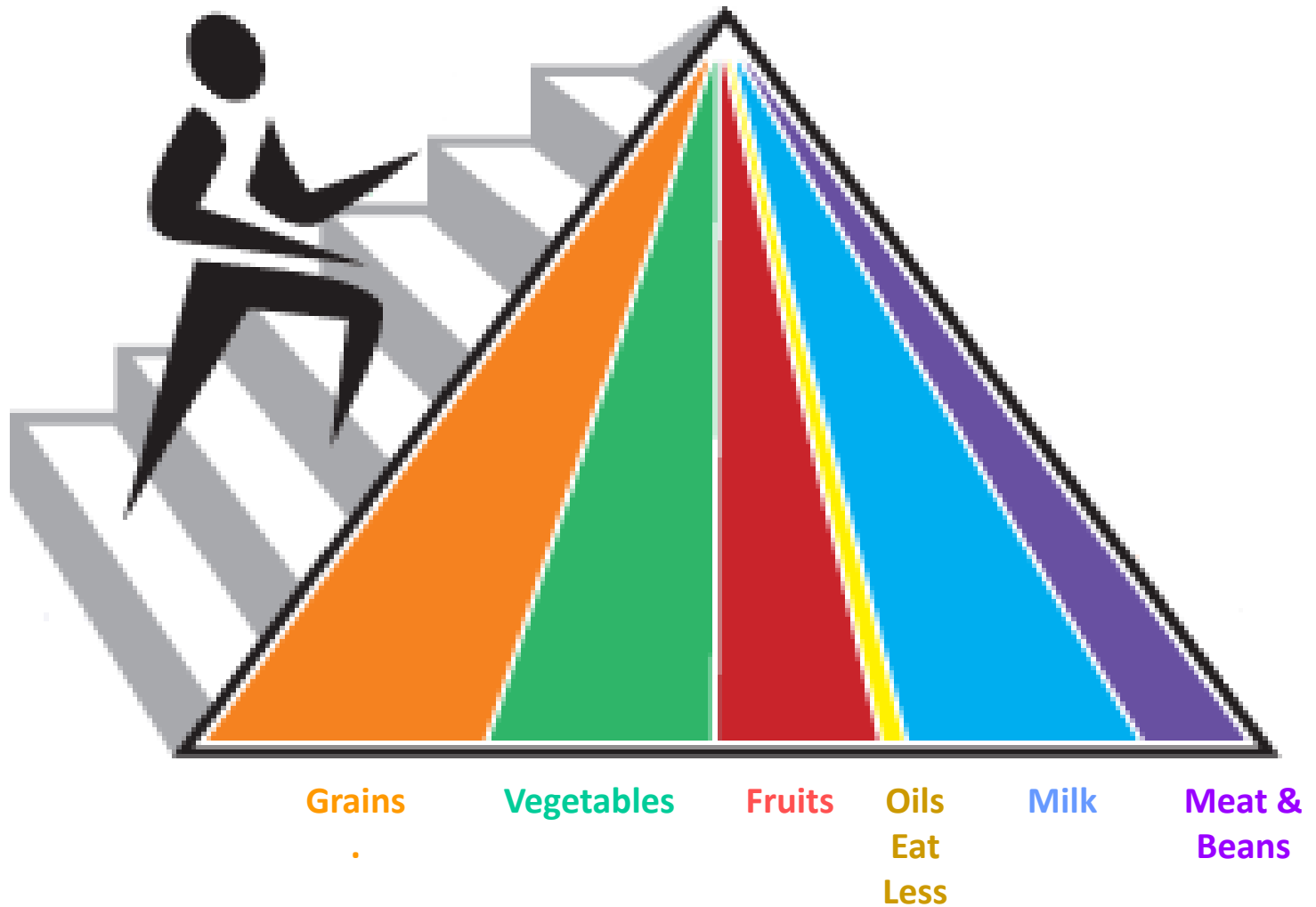
- The direct methods deal
- While indirect methods use

FOOD PYRAMIDS

Food Guide Pyramid



MY Pyramid



My Plates



THE LIFE-COURSE APPROACH TO NUTRITION AND HEALTH

- Nutritional needs should be met at every stage of the life cycle
- In pregnancy →

- Healthy individuals require the same nutrients through-out life,
- There is NO one best diet for everyone.

